Cheng-Yu Liu

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EDUCATION

Department of Physics, National Taiwan University — Taipei,Taiwan
Master of Science — Sept. 2021—July. 2024

Department of Physics, National Central University — Taoyaun, Taiwan
Bachelor of Science — Sept. 2017—June. 2021

EXPERIENCE

Research Assistant — Taipei,Taiwan
Institute of Atomic and Molecular Science, Academic Sinica — Oct. 2024—Presnt

Research Assistant — Taipei,Taiwan
Department of Physics, National Taiwan University — Sept. 2024—Oct. 2024

RELEVANT COURSES

- Introduction to Quantum Optics GPA 4.3/4.3
- Applications of Quantum Computation GPA 4.3/4.3
- Introduction to Quantum Computation and Quantum Information GPA 4.3/4.3
- General Physics A (first semester) 97/100
- Applied Mathematics 97/100
- Mathematical Method in Physics I 99/100
- Quantum and Statistical Physics 94/100
- General Physics A (second semester) 100/100

PUBLICATIONS AND PRESENTATIONS

1. C. G. Feyisa, **Cheng-Yu Liu**, Muhammad S. Hasan, J. S. You, Huan-Yu Ku, and H. H. Jen, *"Robustness of tripartite entangled states in passive PT-symmetric qubits,"* **arXiv:2411.17457** under review (Commun. Phys).
2. **Cheng-Yu Liu**, C. G. Feyisa, Muhammad S. Hasan, and H. H. Jen, *"High-fidelity multipartite entanglement creation in non-Hermitian qubits"* **arXiv:2412.01133** under review (J. Phys. B).
3. **Cheng-Yu Liu**, Ming-Shien Chang, Guin-Dar Lin, *"Novel drive through entangling gate mediated by a cavity"* in preparation will based on my master thesis **https://tdr.lib.ntu.edu.tw/jspui/handle/123456789/94396**. → The work is presented by my Professor Guin-Dar Lin in Joint Symposium on Quantum Computing 2024.

RESEARCH INTERESTS

- Trapped ion system

  Previous work: 1D,2D,3D crystal trajectories mathematical and numerical simulation, geometric phase gate schemes, ion transportation method, suitable species choices.
  My new ideas: 1. A CZ gate based on triggering ions motion when one ion moving inside cavity. 2. Hybird system of ions and atoms. 3. Kicking ions outside Lamb-Dicke limit for distant, scalable connection.
- Rydberg atom platform

  Previous work: Numerical simulation, laser implementation, possible gate schemes, ion transportation method, suitable species choices.
  My new ideas: 1. A possible scheme to connect atoms outside one Rydberg blockade radius. 2. Hybird system of ions and atoms.
- Non-Hermiticity in superconducting qubits

  Previous work: 3-qubit and 4-qubit gate design in non-Hermitain all-to-all connected superconducting qubits, accelerating entanglement compared with Hermitian regime, Pulse shaping in non-Hermitian system.
- Cavity QED system

  Previous work: Gate design utilising spatial dependent Tavis-Cummings model.
  My new ideas: 1. Combination with trapped ion normal modes and cavity mode for a CZ gate. → Analysis of coupling between phonon and photon might be interesting.
- Quantum computing and quantum information

  Previous work: Decomposing an entangled state using an entangling gate, which I designed in a superconducting qubits non-Hermitian system. This gate functions as a combination of two consecutive controlled-controlled-Z gates.

TECHINICAL SKILLS

Numerical calculations and analytical calculations tool, mainly Julia and Mathematica

Experience and ability to pictorize and visulize physics problem and later try to demonstrate in math and coding, especially in the Quantum optics and Quantum information field.

ACTIVITIES

Poster session in annual meeting of the physical society of Taiwan 2024

Poster session and oral presentation in physics annual meeting, National Central University 2020

Participant in Joint Symposium on Quantum Computing 2024

Participant in Workshop on Quantum Science and Technology 2024

Participant in Atomic, Molecular and Optical Summer school 2023

Speaker and vice president in English conversation group in National Central University 2020-2021

Volunteer singer on Taipei Street and member of various Music Clubs